



Review

Vibrations of straight and curved composite beams: A review

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ABSTRACT

Laminated composite straight and curved beams are frequently used in various engineering applications. This work attempts to review most of the research done in recent years (1989–2012) on the vibration analysis of composite beams. This review is conducted with emphasis given to the theory being applied (thin, thick, layerwise), methods for solving equations (finite element analysis, differential transform and others) experimental methods, smart beams (piezoelectric or shape memory), complicating effects in both material and structure (viscoelastic, rotating, tip mass and others) and other areas that have been considered in research. A simple classic and shear deformation model would be explained that can be used for beams with any laminate.

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1. Introduction

Laminated composite beams, plates and shells have been used in extensive applications of many engineering fields in recent decades. Structures composed of composite materials offer lower weight and higher strength and stiffness than those composed of most metallic materials. These advantages coupled with the ability to tailor designs for specific purposes, gave them a competitive edge when compared with normal engineering materials and led to their extensive use. Composite beams, plates and/or shell components have found increasing use in recent aerospace, submarine automotive structures.

One of the important problems in composite structural design is the vibration analysis of composite beams. Composite beams act as lightweight load carrying structures in diverse application.

Literature on composite beam research can be found in many conferences and journals. Kapania and Raciti [1] made a review on advances in analysis of laminated beams and plates vibration and wave propagation in 1989. Rosen [2] reviewed the research on static, dynamic, and stability analysis of pretwisted rods and beams in 1991. Chidamparam and Leissa [3] reviewed the published literature on the vibrations of curved bars, beams, rings and arches of arbitrary shape which lie in a plane in 1993. Also a book [4] was dedicated to vibration of composite beams, plates and shells.

This article focuses on the last two decades of research (1989–2012) done on the vibration analysis of composite beams. The literature is reviewed while focusing on various aspects of research. We will first review the various beam theories that are being used in research in recent years. These include thin (or classical), thick (or shear deformation), and layerwise beam theories. Then different methods for solving equations of motion such as transfer matrix method, finite element method and others would be reviewed. Another aspect of research will be use of smart materials, which include electrorheological fluids, piezoelectric sensors and actuators, and shape memory materials. Complicating effect will be the final category that will be addressed. This will include viscoelastic effects, added mass, rotating beam, beams with damages and so on.

2. Beam theories

Beams are generally three dimensional (3D) bodies bounded by four, relatively close surfaces. The 3D equations of elasticity are unnecessarily complicated when written for a beam. Researchers simplify such equations by making certain assumptions for particular applications. Almost all beam theories reduce the 3D elasticity problem into a one dimensional (1D) problem.

There are two issues typically treated for 1D analysis of beams. The first problem is the issue of coupling and how to include the various couplings (stretching bending, bending twisting and others) that are ignored in reducing 3D equations to 1D. A suitable approach for inclusion of coupling parameters is to redefine stiffness parameters such that it includes other couplings. Coupling problem can be solved by using equivalent stiffness parameters instead of normal definition of A_{11} , B_{11} , and D_{11} . That is why different definitions of stiffness parameters are presented first.

2.1. Stiffness parameters

Kaw [5] suggested to define the stiffness parameters based on the flexibility matrix (the ABD matrix inverse). Rios and Chan [6] proposed another formulation based on ABD inverse matrix. Ecsedi and Dluhi [7] analyzed the static and dynamic behavior of nonhomogeneous curved beams and closed rings. Instead of ABD terms, the modulus was treated as a function in cylindrical coordinates. Some other researchers [8–11] treated the problem of laminated composite beams to that of isotropic homogeneous beams with effective bending (EI), torsional (GJ), and axial stiffness parameters (EA). To the knowledge of authors, none of these models were accurate in analysis of generally laminated beams. Hajianmaleki and Qatu [12] showed that using equivalent modulus of elasticity of each lamina, one can get accurate results for static and dynamic analyses of generally laminated beams with any kind of coupling. The equivalent modulus of elasticity of each lamina is found using [12,13]

$$\frac{1}{E_x^k} = \frac{\cos^4(\theta_k)}{E_{11}} + \left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_{11}} \right) \cos^2(\theta_k) \sin^2(\theta_k) + \frac{\sin^4(\theta_k)}{E_{22}} \quad (1)$$

Equivalent A_{11} , B_{11} and D_{11} using the following equations were suggested for a curved beam [14]

$$\begin{aligned} A_{11} &= R \sum_{k=1}^N b E_x^{(k)} \ln \left(\frac{R+z_k}{R+z_{k-1}} \right) \\ B_{11} &= R \sum_{k=1}^N b E_x^{(k)} \left[(z_k - z_{k-1}) - R \ln \left(\frac{R+z_k}{R+z_{k-1}} \right) \right] \\ D_{11} &= R \sum_{k=1}^N b E_x^{(k)} \left[\frac{1}{2} ((R+z_k)^2 - (R+z_{k-1})^2) - 2R(z_k - z_{k-1}) \right. \\ &\quad \left. + R^2 \ln \left(\frac{R+z_k}{R+z_{k-1}} \right) \right] \end{aligned} \quad (2)$$

and for a straight beam, they reduce to [15,16]

$$\begin{aligned} A_{11} &= \sum_{k=1}^N b E_x^k (h_k - h_{k-1}) \\ B_{11} &= \sum_{k=1}^N b E_x^k \frac{(h_k^2 - h_{k-1}^2)}{2} \\ D_{11} &= \sum_{k=1}^N b E_x^k \frac{(h_k^3 - h_{k-1}^3)}{3} \end{aligned} \quad (3)$$

2.2. Effect of shear deformation

The inclusion of shear deformation in the analysis of beams was first made by Timoshenko [17]. Hence, theories considering shear deformation are called as Timoshenko beam theories. In this regard, the beam theories are classified based on the order of polynomial for approximation of displacements through the thickness. Fig. 1 shows a free body diagram of a differential beam element. Suppose that the displacement u can be expressed as

$$u = u_0 + z \left[c_0 \frac{\partial w}{\partial x} + c_1 \phi(x, t) \right] + c_2 z^2 \psi(x, t) + c_3 \left(\frac{z}{h} \right)^3 \left(\frac{\partial w}{\partial x} + \phi(x, t) \right) \quad (4)$$

where u , and w are displacements in x and z directions, respectively. where ψ , ϕ are the rotation of a line element perpendicular to the original direction in the x and y direction, respectively. For this equation, special cases are defined as [18]

The classical beam theory (CBT): $c_0 = -1$, $c_1 = 0$, $c_2 = 0$, $c_3 = 0$.

The first order shear deformation theory (FSDT): $c_0 = 0$, $c_1 = 1$, $c_2 = 0$, $c_3 = 0$.

The second order shear deformation theory (SSDT): $c_0 = 0$, $c_1 = 1$, $c_2 = 1$, $c_3 = 0$.

The third order shear deformation theory (TSDT): $c_0 = 0$, $c_1 = 1$, $c_2 = 0$, $c_3 = -(4/3)h$.

In classical theories, shear deformation and rotary inertia terms are ignored and is generally accurate for thin beams; while in shear deformation theories, some of those terms are included and accuracy for thick beams increases. Shear deformation theories with higher order than FSDT are treated in one section in this paper titled higher order shear deformation theories.

2.2.1. Classical beam theory

If the beam thickness is less than 1/20 of the wavelength of the deformation mode, a classical beam theory (CBT) or Euler Bernoulli (EB) beam theory where shear deformation and rotary inertia are negligible, is generally acceptable for lower frequencies determination. Qatu [19] and Qatu and Elsharkawi [20] used CBT to study vibration of straight and curved cross-ply laminated beams. They used Ritz method for solving the equations of motion. Qatu and Iqbal [21] used CBT to solve the vibration of a cross-ply laminated composite driveshaft with an intermediate joint.

Mei [22] studied the effect of coupling between bending and torsional deformations on vibrations of composite EB beams from a wave vibration standpoint. The torsional mode was found affected by the material coupling only at low frequencies. The flexural modes were found to be affected by material coupling over the entire frequency band. Mei [23] also studied the local wave transmission and reflection characteristics at various discontinuities on composite beams. Gunda et al. [24] investigated large amplitude vibration of laminated composite beams with symmetric and asymmetric layup orientations and axially immovable ends. They used CBT and solved the equations by Rayleigh–Ritz method. Geometric nonlinearity of von-Karman type was considered which accounts for the membrane stretching action of the beam.

Hajianmaleki and Qatu [25] used the following stiffness parameters in a FSDT formulation to study transverse vibration of generally laminated two-segment shafts (see Fig. 2).

$$A_{11} = \pi \sum_{k=1}^N E_x^k (r_k^2 - r_{k-1}^2)$$

$$D_{11} = \frac{\pi}{4} \sum_{k=1}^N E_x^k (r_k^4 - r_{k-1}^4) \quad (5)$$

$$B_{11} = \pi R \sum_{k=1}^n \left[c^4 Q_{11} + c^2 s^2 (Q_{12} + 2Q_{66}) + \frac{3}{8} s^4 Q_{22} \right] (z_k^2 - z_{k-1}^2)$$

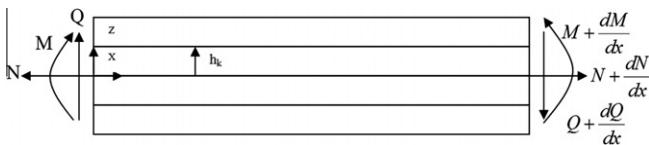


Fig. 1. Parameters of the beam model.

where z in Eq. (5) is measured from middle of the laminate. They used general differential quadrature (GDQ) for solution of the equations and the natural frequencies for two-segment shafts with different laminates and length ratios were assessed.

In the CBT approach by Qatu [15], strain and curvature change at the middle surface are:

$$\varepsilon_0 = \frac{\partial u}{\partial x}, \quad \kappa = \frac{\partial^2 w}{\partial x^2} \quad (6)$$

Normal strain at any point would be

$$\varepsilon = \varepsilon_0 + z\kappa \quad (7)$$

Force and moment resultants are calculated using

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A_{11} & B_{11} \\ B_{11} & D_{11} \end{bmatrix} \begin{bmatrix} \varepsilon_0 \\ \kappa \end{bmatrix} \quad (8)$$

The equations of motion are

$$\frac{\partial^2 M}{\partial x^2} = I_1 \frac{\partial^2 w}{\partial t^2} - p_z \quad (9)$$

$$\frac{\partial N}{\partial x} = I_1 \frac{\partial^2 u}{\partial t^2} - p_x \quad (10)$$

where I_1 , p_x , and p_z are beam mass, external loads in x and z directions per unit length, respectively.

$$I_1 = \sum_{k=1}^N b \rho^{(k)} (h_k - h_{k-1}) \quad (11)$$

Inserting the strain and curvature relations in the force and moment resultants equations and using those in the equations of motion, one can express the equations of motion in terms of displacements. Applying boundary condition and suitable functions for displacements will result in characteristic equation for natural frequencies of the beam.

2.2.2. Shear deformation theories

Suresh et al. [26] investigated the effect of warping on the free vibration of torsional–flexural coupled beams. Abramovich [27] studied free vibration of symmetric composite beams with a term representing the joint action of shear deformation and rotary inertia that was omitted in the Timoshenko equations. Song and Librescu investigated the free vibration behavior of composite

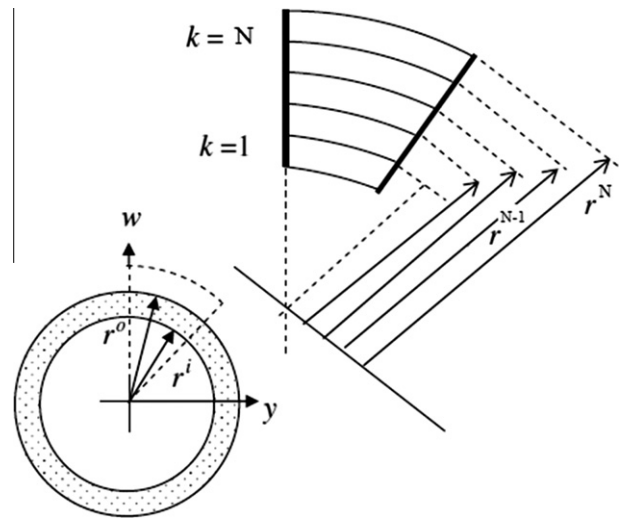


Fig. 2. Geometry of the shaft model.

thick- and thin-walled, single-cell closed [28] and open [29] beams of arbitrary cross-section. Cortinez and Piovan [30] developed a theoretical model for the dynamic analysis of composite thin-walled beams with open or closed cross-sections. Their model incorporated the shear flexibility as well as a state of initial stresses. Lee et al. [31] derived differential equations governing the free vibrations of elastic, horizontally curved beams with unsymmetric axes in cartesian coordinates, including the effect of torsional inertia. They numerically computed the frequencies and mode shapes for parabolic curved beams with both clamped ends and both hinged ends. In another research [32] they included the effects of axial extension, shear deformation, and rotatory inertia. Karama et al. [33] proposed a multi-layer laminated composite structure model to predict the mechanical behavior of multi-layered laminated composite structures using exponential function as a shear stress function. They validated the model for different cases in bending, buckling and free vibration on a cross-ply laminate. Kim et al. [34] studied the effects of the steel core or casing on the bending natural frequency of composite shafts for simply supported boundary conditions.

Mei [35] presented the effect of coupling between bending and torsional deformations on vibrations of symmetric composite Timoshenko beams from a wave vibration standpoint. He found that the torsional modes at low frequencies and cutoff transitional frequency are unaffected by material coupling. Mei [36] also presented wave vibration analysis of axially loaded bending-torsion coupled composite Timoshenko beam structures.

2.2.3. First order shear deformation theories

FSDT methods have been used by many researchers. The works by Chandrashekhara et al. [37], Krishnaswamy et al. [38], Abramovich et al. [39,40] was validated for symmetric cross-ply laminates that have no coupling. In the FSDT model by Teboub and Hajela [41] symmetric beams having fibers in one direction were considered. Bert and Kim [42] proposed a FSDT for predicting the critical speed of a shear deformable, composite driveshaft. They modeled the shaft as a Bresse–Timoshenko beam (FSDT with rotary inertia and gyroscopic action) generalized to include bending-twisting coupling. The FSDT models by Eisenberger et al. [43] and Qatu [15] for curved beams were also validated for cross-ply laminates. In FSDT approach by Qatu [15], strains and curvature changes at the middle surface are:

$$\varepsilon_0 = \frac{\partial u}{\partial x}, \quad \kappa = \frac{\partial \psi}{\partial x}, \quad \gamma = \frac{\partial w}{\partial x} + \psi \quad (12)$$

where γ is the shear strain at the neutral axis.

Force and moment resultants as well as shear forces are calculated using

$$\begin{bmatrix} N \\ M \\ Q \end{bmatrix} = \begin{bmatrix} A_{11} & B_{11} & 0 \\ B_{11} & D_{11} & 0 \\ 0 & 0 & A_{55} \end{bmatrix} \begin{bmatrix} \varepsilon_0 \\ \kappa \\ \gamma \end{bmatrix} \quad (13)$$

where A_{55} is [12]

$$A_{55} = \frac{5}{4} \sum_{k=1}^N b \bar{Q}_{55}^k \left[(h_k - h_{k-1}) - \frac{4}{3h^2} (h_k^3 - h_{k-1}^3) \right], \quad (14)$$

The equations of motion considering rotary inertia and shear deformation would be

$$\begin{aligned} \frac{\partial N}{\partial x} &= I_1 \frac{\partial^2 u}{\partial t^2} + I_2 \frac{\partial^2 \psi}{\partial t^2} - p_x \\ -\frac{\partial Q}{\partial x} &= p_z - I_1 \frac{\partial^2 w}{\partial t^2} \end{aligned} \quad (15)$$

$$\frac{\partial M}{\partial x} - Q = I_2 \frac{\partial^2 u}{\partial t^2} + I_3 \frac{\partial^2 \psi}{\partial t^2}$$

$$(I_1, I_2, I_3) = \sum_{k=1}^N b \rho^{(k)} ((h_k - h_{k-1}), \frac{1}{2} (h_k^2 - h_{k-1}^2), \frac{1}{3} (h_k^3 - h_{k-1}^3))$$

Further mathematical manipulation enables one to express the equations of motion in terms of displacements. Applying boundary conditions and suitable functions for displacements will result in characteristic equation for natural frequencies of the beam. Hajianmaleki and Qatu [44] showed using FSDT approach for thick beams along with equivalent modulus of elasticity for calculation of ABD parameters (Eqs. (2) and (3)) one can reach accurate static and vibration analysis of composite straight beams of all laminates. They verified their model comparing to 3D finite element method (FEM) using ANSYS software.

They showed that [14] by exact application of deepness term $(1 + z/R)$ in derivation of ABD parameters, FSDT can accurately analyze the static and vibration behavior of laminated curved beams with any lamination, deepness, and boundary conditions. They found exact solution for simply supported and numerical solutions for other boundary conditions.

Exact solution is available for the simply supported boundary conditions where we can assume

$$[u_0, w_0, \psi] = \sum_{m=1}^m [A_m \cos(\alpha_m x), C_m \sin(\alpha_m x), B_m \cos(\alpha_m x)] \sin(\omega t) \quad (16)$$

where $\alpha_m = m\pi/a$ and a is the beam length. For other boundary conditions, the analysis is not this straightforward and a numerical solution is usually employed.

For the simply supported boundary condition, the loading is expanded using Fourier transform

$$[p_x, p_z] = \sum_{m=1}^M [p_{xm} \cos(\alpha_m x), p_{zm} \sin(\alpha_m x)] \quad (17)$$

Substituting these equations in the equations of motion, the solution for free vibration analysis

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{12} & C_{22} & C_{23} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} \begin{bmatrix} A_m \\ C_m \\ B_m \end{bmatrix} + \omega^2 \begin{bmatrix} \bar{I}_1 & 0 & \bar{I}_2 \\ 0 & -\bar{I}_1 & 0 \\ \bar{I}_2 & 0 & \bar{I}_3 \end{bmatrix} \begin{bmatrix} A_m \\ C_m \\ B_m \end{bmatrix} = 0 \quad (18)$$

where

$$C_{11} = -\alpha_m^2 A_{11} - A_{55}/R^2, C_{22} = \alpha_m^2 A_{55} + A_{11}/R^2, C_{33} = -\alpha_m^2 D_{11} - A_{55}$$

$$C_{21} = -C_{12} = \alpha_m \left[\left(\frac{A_{11} + A_{55}}{R} \right) \right] C_{31} = C_{13} = -\alpha_m^2 B_{11} + A_{55}/R^2$$

$$C_{23} = -C_{32} = \alpha_m \left[\left(\frac{B_{11}}{R} \right) - A_{55} \right]$$

The nontrivial solution for natural frequencies can be found by setting the determinant of characteristic equation matrix to zero. Table 1 summarizes results obtained by the above equations and shows that these equations do indeed provide an accurate shear deformation theory for thick curved beams that are generally laminated. The FSDT results provided by new stiffness parameters (column with FSDT (E_x)) title is shown to be the most accurate compared to 3D FEM simulations.

2.2.4. Higher order shear deformation theories

Some researchers developed HSDT methods for analysis of composite beams. Carrera and Petrolo [45] worked on the effectiveness of higher-order terms in refined beam theories. They concluded that the kinematics model that suits specific problem is deter-

mined by the cross-section geometry and the loading case. Khدير and Reddy [18] determined natural frequencies of the third-order, second-order, first-order and classical arch theories for cross-ply laminates. In another research, they developed analytical solutions of refined beam theories to study the free vibration behavior of cross-ply rectangular beams with arbitrary boundary conditions in conjunction with the state space approach. They showed that the disagreement between different shear deformation theories is much less than the disagreement between any of them and EB theory [46].

Suresh and Nagaraj [47] proposed a HSDT for the static and dynamic analysis of thin-walled composite beams with open and closed cross sections. Their method was validated by comparison to analytical and experimental results of static deflections of composite beams with symmetric and antisymmetric lay-ups. Song and Waas [48] studied buckling and free vibration of stepped laminated composite beams using HSDT assuming a cubic distribution for the displacement field through the thickness. The results were compared to CBT and FSDT and did not show significant differences to those for a wide range of beam geometry and material properties. Marur and Kant [49] proposed three higher order refined displacement models for the free vibration analysis of sandwich and composite beams. All higher order models were found to compute frequencies which were numerically higher than those of FSDT for the thin beams considered. In the case of thick sandwiches, higher order theories give quite significantly lower frequencies than Timoshenko theory. Kant et al. [50] proposed an analytical higher order model using C^0 continuous functions for symmetric laminates. Matsunaga [51] studied vibration and buckling of cross-ply laminated beams according to HSDT. Subramanian [52] proposed two higher order and two FEM approaches and validated them for symmetric laminated beams and different slenderness ratios.

Machado et al. [53] investigated dynamic stability of thin-walled composite beams, considering shear deformation, subjected to axial external force. They used Galerkin's method in order to discretize the governing equation and the Bolotin's method to determine the regions of dynamic instability of a simply supported beam. The numerical results showed that longitudinal vibration has large influence when the forcing frequency approaches the natural longitudinal frequency, obtaining wider parametric instability regions. Machado and Cortinez [54] showed that when the ratio of the smaller axis to the major axis flexural stiffness is large, classic analysis of vibration may lead to inaccurate predictions because of the effects of initial displacements.

Zhen and Wanji [55] compared different displacement-based theories for vibration of special symmetric and anti-symmetric composite and sandwich beams including zigzag model and different HSDTs. They proposed the global–local HSDT suitable for symmetric and anti-symmetric laminates. Emam and Nayfeh [56] proposed a closed-form solution for the postbuckling deformation as a function of the applied axial load, which is beyond the critical buckling load. They exactly solved the linear vibration problem around the first buckled configuration. El Fatmi and Ghazouani [57] proposed a HSDT based on a kinematics built from the exact form of Saint–Venant displacement for symmetric cross-sections. Giunta et al. [58] found torsional, axial, shear and mixed mode natural frequencies using refined theories. The 3D displacement field was approximated as a generic N-order polynomial expansion such that several HSDT and CBT models accounting for non-classical effects could be formulated.

2.3. Layerwise theories

In general, layerwise laminate theories are used to represent local effects more accurately, such as interlaminar stress distribution, and delamination. These theories are typically employed for cases involving anisotropic materials in which transverse shear effects cannot be ignored. Singh and Gupta [59] presented formulations for the dynamic analysis of a composite rotors by using conventional equivalent modulus beam theory (EMBT) and a layerwise beam theory. In layerwise theory, the displacement function u for a point of the i th lamina at a distance z_i from the midplane is

$$\bar{u}_{zi} = \frac{\bar{u}_{i+1} + \bar{u}_i}{2} + z_i \frac{\bar{u}_{i+1} - \bar{u}_i}{h_i} \quad (19)$$

The results indicated that the difference between the two theories was not large, but for unsymmetric stacking sequences with bending stretching couplings, the EMBT may result in inaccurate predictions of rotor dynamic behavior. Lee [60] used a layerwise theory for free vibration analysis of a laminated beam with delamination.

2.4. Other theories

Pai and Nayfeh [61] proposed nonlinear equations describing the extensional–flexural–flexural–torsional vibrations of slewing or rotating metallic and composite beams. Three consecutive Euler angles were used to relate the deformed and undeformed states.

Table 1
Nondimensional natural frequencies $\Omega = \omega a^2 \sqrt{12\rho/E_1 h^2}$ of simply supported curved beam.

n	$a/R = 0.2$				$a/R = 1$				$a/R = 2$			
	CBT	FSDT (Q_{11})	FSDT (E_x)	3D FEM	CBT	FSDT (Q_{11})	FSDT (E_x)	3D FEM	CBT	FSDT (Q_{11})	FSDT (E_x)	3D FEM
$[0_2/90_2]$												
1	4.638	4.584	4.611	4.596	3.933	3.890	4.040	4.019	2.278	2.254	2.399	2.377
2	18.61	17.78	17.88	17.64	17.63	16.88	17.53	17.27	15.40	14.78	15.91	15.61
3	41.77	37.97	38.16	37.16	40.38	36.83	38.13	37.06	37.61	34.45	36.93	35.78
4	73.98	63.28	63.56	61.02	72.03	61.92	63.84	61.20	68.58	59.31	63.13	60.35
5	115.0	92.07	92.42	87.58	112.3	90.53	92.98	87.98	108.1	87.74	92.69	88.33
$[45]_4$												
1	5.627	5.532	3.283	3.331	4.848	4.769	2.830	2.871	2.841	2.796	1.659	1.682
2	22.61	21.20	12.94	12.94	21.80	20.45	12.48	12.51	19.39	18.21	11.12	11.17
3	50.92	44.53	28.29	28.24	50.10	43.83	27.84	27.82	47.60	41.68	26.48	26.52
4	90.55	72.90	48.39	48.13	89.72	72.25	47.97	47.72	87.19	70.26	46.67	46.48
5	141.5	104.2	72.30	71.54	140.7	103.6	71.90	71.16	138.1	101.8	70.67	70.00
$[30_2/60_2]$												
1	5.127	5.054	3.402	3.463	4.463	4.400	2.951	3.002	2.634	2.598	1.738	1.765
2	20.58	19.49	13.39	13.43	20.07	18.99	13.00	13.07	18.08	17.10	11.67	11.75
3	46.28	41.30	29.20	29.34	46.06	41.01	28.93	28.96	44.37	39.43	27.73	27.83
4	82.13	68.23	49.82	49.68	82.33	68.16	49.69	49.50	81.16	66.93	48.70	48.52
5	128.0	98.42	74.23	73.58	128.7	98.53	74.23	73.53	128.3	97.61	73.46	72.74

The nonlinear equations of motion were used to investigate the response of an inextensional, symmetric angle-ply beam to a harmonic base-excitation along the flapwise direction [62] and forced nonlinear vibration of a symmetrically laminated beam [63]. Kapuria et al. [64] used zigzag theory to satisfy continuity of transverse shear stress through the laminate to assess the dynamic and buckling response of laminated beams. Kovacs [65] proposed an iterative laminate model that can accurately determine the dynamic stress distribution in soft and hard cored sandwich arches.

Lee [66] applied pseudospectral method to the free vibration analysis of circularly curved multi-span Timoshenko beams and computed natural frequencies in good agreement with the literature. Sapountzakis and Dourakopoulos [67,68] developed a boundary element method (BEM) for the general flexural–torsional vibration problem of Timoshenko beams of arbitrarily shaped composite cross-section taking into account the effects of warping stiffness, warping and rotary inertia and shear deformation. The BEM required only boundary discretization resulting in line or parabolic elements instead of area elements of the FEM solutions.

3. Methods for solving equations of motion

3.1. Differential transform method

The differential transform method (DTM) is a semi-analytic transformation technique based on the Taylor series expansion and is a useful tool to obtain analytical solutions of the differential equations. In this method, certain transformation rules are applied and the governing differential equations and the boundary conditions of the system are transformed into a set of algebraic equations in terms of the differential transforms of the original functions and the solution of these algebraic equations gives the desired solution of the problem. Ozgumus and Kaya [69] introduced the DTM to study the vibration characteristics of a rotating tapered cantilever EB beam with linearly varying rectangular cross-section of area proportional to x^n . In other researches they studied the out-of-plane free vibration analysis of a double tapered EB beam, mounted on the periphery of a rotating rigid hub [70] and performed free vibration analysis of a rotating, double tapered Timoshenko beam featuring coupling between flapwise bending and torsional vibrations using DTM [71]. Kaya and Ozgumus [72,73] analyzed the free vibration response of an axially loaded closed-section composite Timoshenko beam which featured material coupling between flapwise bending and torsional vibrations using DTM.

Other people who used DTM were Catal [74] who solved differential equations of motion for free vibration of axially loaded beam supported on elastic soil and Arikoglu and Ozkol [75] who made vibration analysis of composite sandwich beams with viscoelastic core.

3.2. Dynamic stiffness matrix method

Abramovich et al. [76] proposed the exact element method to find vibration frequencies of multi-span laminated beams, including the effect of rotary inertia and shear deformation. They derived the dynamic stiffness matrix to solve for any set of boundary conditions including elastic connections, number and length of spans, as in the classical direct stiffness method for framed structures. They validated their model for symmetric cross-ply laminates. Banerjee and Williams [77,78] used dynamic stiffness matrix in conjunction with the Wittrick–Williams algorithm to compute the natural frequencies and mode shapes of composite beams with substantial coupling between bending and torsional displacements. Banerjee [79] studied free vibration analysis of axially

loaded composite Timoshenko beams by using the dynamic stiffness matrix method. Banerjee and Su [80] developed the dynamic stiffness matrix of a thin-walled spinning composite beam and investigated its free vibration characteristics based on CBT. Tseng et al. and Huang et al. [81,82] developed an analytical solution for the free vibration of composite laminated beams of variable curvature using Timoshenko-type curved beam theory and incorporating the dynamic stiffness method.

Tseng et al. [83] combined the dynamic stiffness method with the Laplace transform to obtain accurate transient responses of an arch with variable curvature. They considered the effects of shear deformation, rotary inertia, and damping. Huang et al. [84] developed dynamic stiffness matrix for noncircular curved beams with variable cross-section to derive an exact solution of the out-of-plane free vibration. Jun et al. [85] used dynamic stiffness approach to numerically investigate vibration of laminated composite beams with arbitrary ply orientation. They accounted for the influences of Poisson effect, shear deformation and rotary inertia. They included the effect of axial load in another research [86]. Moon-Young et al. [87] proposed the exact dynamic and static stiffness matrices of shear deformable non-symmetric thin-walled beam-columns. They found the total potential energy in the general form by introducing the displacement field based on semitangential rotations and deriving transformation equations between displacement and force parameters defined at the arbitrary axis and the centroid-shear center axis, respectively.

3.3. State space approach (transfer matrix method)

The state space approach or transfer matrix method (TMM) in vibration analysis of beams was first used by Khdeir and Reddy [88]. They studied free vibration of cross-ply laminated beams with arbitrary boundary conditions and presented the fundamental natural frequencies based on the various beam theories. Yildirim et al. [89] studied the in-plane free vibration problem of symmetric cross-ply laminated beams based on the TMM. They compared the results of Timoshenko and EB theories for first six non-dimensional frequencies with each other for length-to-thickness ratios. They included out of plane vibration in another research [90]. Yildirim [91] studied axial and shear deformation effects on the in-plane natural frequencies of symmetric cross-ply laminated circular arches and obtained the exact in-plane element stiffness matrix based on the TMM. In another research, the effect of the longitudinal to transverse moduli ratio on the first three in-plane natural frequencies for different length/thickness ratios and boundary conditions of symmetric cross-ply beams was investigated [92]. Yildirim and Kiral [93] studied the effect of shear deformation and rotary inertia on out-of-plane free vibration problem of symmetric cross-ply laminated beams by the TMM. Yildirim [94] also did a numerical study to investigate the common effects of the rotary inertia and shear deformation on the first six out-of-plane free vibration frequencies of symmetric cross-ply laminated bars with the help of TMM.

3.4. Finite element methods

FEM is used in solving equations of motion of composite beams mainly for incorporation of nonlinear effects and complicated geometries such as turbopropellers [95], helically wound symmetrical tubes [96], delamination length and location [97] tapered beams [98], and conical tubes [99].

Different beam theories such as equivalent isotropic behavior [100], and CBT [101,102] were used in implementation of FEM. Other theories such as FSDT was used for bi-axial bending as well as torsion [103], curved beams [104], vibration and wave propagation analysis [105], thin walled beam of arbitrary cross-section

[106], generally laminated beams [107]. HSDT was used for general beams [108,109], damped delaminated beam [110], axial–flexural–shear coupled deformation [111], and tapered beams [112,113]. Nonlinear theories were used for general beams [114,115], thin-walled curved beams with non-symmetric open cross section [116] resting on a class of two parameter elastic foundation [117].

Hodges et al. [118] compared different methods for determination of cross sectional stiffness parameters and solved the equations of motion for eigenvalues using numerical integration and mixed FEM. Both of these methods were validated for symmetric beams. Lee et al. [119] proposed a FEM for stress and vibration analysis of laminated composite beams based on a multilayered theory. Their theory accounted for the continuity of interlaminar shear stress.

Zeng [120] developed a beam element for the Composite Element Method (CEM) where both FEM and classic theory were considered. Zapfe and Lesieutre [121] proposed a discrete layer beam FEM for the dynamic analysis of composite sandwich beams with integral damping layers. Yanchu [122] decomposed the laminate as multiple basic layers and assembled mass and stiffness matrices of these basic layers together to generate those matrices for the beam element. His method allowed the direct consideration of complex modulus of any layer so that it offered much more accurate damping analysis for such structures. Kapania and Goyal [123] developed three models to predict randomness in the free vibration response of unsymmetrically laminated beams: exact Monte Carlo simulation, sensitivity-based Monte Carlo simulation, and probabilistic FEA. Results showed that variations of 5° in ply angles have little effect on the lower mode natural frequencies of unsymmetrically and symmetrically laminated beams. Piovan and Cortinez [124] carried out parametric studies of the natural frequencies of tailored composite thin-walled curved box-beams by means of FEM. Their structural model took into account the shear flexibility due to warping as well as due to bending.

Kim et al. [125] proposed a formulation for free vibration and spatial stability of non-symmetric thin-walled curved beams considering variable curvature effects and the second-order terms of finite semitangential rotations. They developed a thin-walled curved beam element using the third-order Hermitian polynomials and compared the numerical solutions with the results analyzed by ABAQUS shell elements.

Sarikanat et al. [126] determined the effects of axial load on the natural frequency of simply supported thick composite beams using FEM. The material properties of the elements were calculated with two different average value methods (arithmetic and weighted average). They observed that the results obtained by the arithmetic average method were quite close to the analytic results. Saravanos et al. [127] developed a 3D shear beam FEM for the damping analysis of tubular laminated composite beams. Sapountzakis and Mokos [128] developed a boundary element method to perform dynamic analysis of 3D composite beam elements subjected in arbitrarily distributed dynamic loading. Piovan and Cortinez developed a theoretical model for the generalized linear analysis of composite thin-walled straight [129] and curved [130] beams with open or closed cross-sections incorporating full form of the shear deformability. They used the theoretical formulation together with a non-locking fourteen DOF FEM for the solutions to the general equations of thin-walled shear deformable composite beams.

Boukhalifa et al. [131] employed a p-version, hierarchical FEM to investigate dynamic behavior of the rotating composite shaft on rigid bearings. They incorporated the transverse shear deformation, rotary inertia and gyroscopic effects, as well as the coupling effect. Kadioglu and Iyidogan [132] investigated free vibration of laminated composite curved beams using mixed FEM. Vo and Lee [133] developed a general analytical model based on SDBT to study

the flexural–torsional coupled vibration and buckling of thin-walled open section symmetric laminated beams with arbitrary lay-ups using FEM. They studied effect of constant axial force using CBT [134] and SDBT [135]. In another research, they developed a 1D FEM based on CBT with seven DOFs per node to present the interaction curves for vibration and buckling of thin-walled composite box beams under constant axial loads and equal end moments [136]. Vo et al. [137] studied vibration of thin-walled composite beams with I-shaped cross-sections. Kim and Wang [138] carried out vibration analysis of composite beams by using FEM-based formal asymptotic expansion method. They used 3D equilibrium equations in which cross-sectional coordinates were scaled by the characteristic length of the beam and discretized microscopic two-dimensional (2D) and macroscopic 1D equations obtained via the asymptotic expansion by applying a conventional FEM. Slimani et al. [139] characterized the free nonlinear vibration behavior of composite beams by using polynomial FEM with shape functions based on Legendre polynomials or sinusoidal functions. They considered the influence of the vibration amplitude on the modal frequencies of the structure.

4. Experimental investigation

Cudney and Inman [140] derived a method of estimating the distributed damping parameters of a beam based on frequency and damping ratios. Three different mathematical models (viscous damping, strain-rate damping, and both) were used to model the damping mechanism of a quasi-isotropic pultruded cantilevered beam. It was found by experimental modal analysis that the two-parameter damping model provides the best fit to measure modal data. Chandra and Chopra [141] presented a theoretical-cum-experimental study of the free vibration characteristics of thin-walled rotating box beams with bending-twist and extension-twist coupling using Galerkin method. The experimental frequencies and mode shapes correlated satisfactorily with the theoretical results. It was shown that bending–shear coupling unlike extension–shear coupling significantly influences the flexural vibration frequencies of antisymmetric box beams. Bassiouni et al. [100] used FEM including shear deformation and compared numerical results for the natural frequencies with the experimental ones. The theoretical model gave good results compared to the experimental ones. Baba and Thoppul [142] made a experimental study to determine the effect of curvatures and debond on the flexural stiffness and strength of composite sandwich beam structures. Ooijevaar et al. [143] used experiments to investigate vibration based damage identification method for a 2.5-dimensional composite T-beam.

5. Smart beams

Composite materials have the advantage of easily placing smart sensors and actuators inside structure. Smart materials such as piezoelectric sensors and actuators, shape memory alloys, and electro-rheological fluids are used in dynamic studies of beams mainly for control of vibration. Choi et al. [144] studied the effect of electro-rheological fluid on the vibration characteristics of a composite beam. They obtained the complex moduli of a hollow beam filled with an electrorheological fluid by analyzing the beam's motion in free oscillation.

5.1. Piezoelectric beams

Piezoelectric actuators are mostly used for control of shape [145] and vibration [146–154] of composite beams. It is mostly used for control of wings and blades in aerospace industries.

Spearritt and Asokanathan [155] proposed a theoretical and experimental development of a laminated spatially distributed piezoelectric torsional vibration actuator for a clamped–free cantilever beam. Their theoretical and experimental decay histories of the first torsion mode indicated a reduction in decay time for the controlled beam of greater than 10 times that of the uncontrolled beam. Chattopadhyay et al. [156] investigated vibration reduction in rotor blades using active composite box beam. They used a 3D model that approximated the elasticity solution so that the beam cross-sectional properties were not reduced to 1D parameters and both in-plane and out-of-plane warpings were included.

Varadarajan et al. [157] developed a FEM based on HSDT, accounting for piezoelectric effects to investigate performance of an LQG/LTR-based multi-input multi-output robust vibration control system for a laminated composite beam. Takawa et al. [158] investigated the fuzzy control of vibration for a hybrid smart CFRP cantilevered beam actuated by piezoceramic and electro-rheological fluids actuators. Abramovich and Livshits [159] developed a balanced model describing the behavior of piezolaminated composite beams based on FSDT. Raja et al. [160] used quasi-static equations of piezoelectricity to derive a FEM capable of modeling two different kinds of piezoelastically induced actuation in a sandwich beam. They developed a control scheme based on the linear quadratic regulator/independent modal space control method and estimated the active stiffness and the active damping introduced by shear and extension-bending actuators. The shear actuator was more efficient in controlling the vibration than the extension-bending actuator for the same control effort.

Waisman and Abramovich [161] investigated the stiffening effects of a simply supported and clamped–free symmetric piezolaminated composite beam. Takawa and Fukuda [162] and Susumu and Takeshi [163] used a fuzzy model to investigate vibration control for a smart CFRP beam actuated by piezoceramic and electro-rheological fluid actuators. Mitra et al. [164] made theoretical (FEM) and experimental investigation of vibration control of composite box beams using distributed, surface mounted piezoelectric actuators. Chandiramani et al. [165] worked on optimal vibration control of a rotating composite beam with distributed piezoelectric sensing and actuation. They modeled a rotating composite blade, as a box-beam with transverse shear flexibility, shear-tractionless bounding faces and restrained warping, and subjected to a time dependent pressure pulse. Librescu and Na [166] studied the vibration control of adaptive doubly-tapered cantilevered composite beams, simulating an aircraft wing, exposed to time-dependent external pulses through the converse piezoelectric effect.

Gu and Song [167] used piezoceramic patch sensors and actuators for active vibration suppression of a composite I-beam using fuzzy positive position control. Ashida et al. [168] investigated control of thermally induced vibration in a composite beam with damping effect. Their beam consisted of a central thermoelastic structural layer and two outer piezothermoelastic layers. Choi et al. [169] studied bending vibration control of the pre-twisted rotating composite thin-walled beam. The formulation was based on single cell composite beam including a warping function, centrifugal force, Coriolis acceleration, pre-twist angle and piezoelectric effect. Fridman and Abramovich [170] used FSDT to compute natural frequencies and their associated mode shapes, as well as, buckling loads for beams with and without piezoelectric layers influence, with various boundary conditions and lay-ups.

Ji et al. [171] proposed an improved semi-active control synchronized switch damping on voltage (SSDV) method and applied to the vibration control of a composite beam. Also Ji et al. [172] proposed an adaptive semi-active SSDV method based on the least mean squares (LMS) algorithm and applied to the vibration control of a composite beam. Susanto [173] presented an analytical model of piezoelectric laminated slightly curved beams, which included

the computation of natural frequencies, mode shapes and transfer function formulation using the distributed transfer function method (DTFM).

Vadiraja and Sahasrabudhe [174] used HSDT for structural modeling of rotating pre-twisted thin-walled composite beams with embedded macro fiber composite actuators and sensors. It was observed that gyroscopic coupling between lagging-extension motions had significant effect and cannot be neglected in the analysis. Foda et al. [175] developed an analytical approach to suppress the steady state transverse vibration of a symmetric cross-ply laminated composite beam that is excited by an external harmonic force by piezoelectric patches. They used dynamic Green's functions to solve governing equations and proposed a scheme for determining the values of the driving voltages, the dimensions of the PZT patches and their locations along the beam. They performed this study with masses and/or translational and rotational grounded springs instead of piezoelectric patches in another research [176]. Chandiramani [177] studied the optimal control of a thin-walled rotating beam using HSDT. The pretwisted, doubly tapered beam was comprised of orthotropic host with surface-embedded transversely isotropic piezoelectric sensor–actuator pairs. Abramovich [178] showed that in-plane cross sectional deformations due to small lateral vibrations, when coupled with the internal stresses due to the constant electric voltage, results in a significant influence on the free lateral vibrations.

5.2. Beams with shape memory alloys

Lau et al. [179] investigated the change of natural frequencies of a clamped–clamped composite beam with embedded shape memory alloy (SMA) wires. Tsai and Chen [180] investigated numerical parametric studies of the natural frequencies and static buckling loads of the composite beam with activated SMA fibers.

Aoki and Shimamoto [181] investigated the active damping effect of the epoxy resin with embedded fibers of TiNi to examine the possibility to apply the composite as a damping material. Zhang and Zhao [182] studied the kinematic assumptions influence on deflection and vibration characteristics of a composite beam with arbitrarily embedded SMA. They compared natural frequencies of the composite beam with the nonlinear governing equation, which were obtained by directly linearizing the equations and locally linearizing the equations around each equilibrium. Majewska et al. [183] investigated active vibration control of a cracked composite beam using magnetic shape memory alloy actuators. Lee et al. [184] used transfer matrix method to study lateral vibration of a composite stepped beam consisted of SMA helical spring based on CBT. A discussion on this paper was made by Sinha [185]. Dos Reis et al. [186] studied vibration attenuation in an epoxy smart composite beam with embedded NiTi shape memory wires.

6. Complicating effects

6.1. Dynamic loading and excitation

Gong and Lam [187] studied transient response of layered composite beams subjected to underwater shock. Ganesan and Kowda [188] investigated free vibration of composite beam-columns with stochastic material and geometric properties subjected to random axial loads. They used the perturbation method in the context of stochastic analysis. Jun and Xianding [189] studied the flexure–torsion coupled random response of composite beams with solid or thin-walled closed sections subjected to various types of concentrated and distributed random excitations. They assumed random excitations to be stationary, ergodic and Gaussian and obtained analytical expressions for the displacement response of

the composite beams by using normal mode superposition method combined with frequency response function method.

Li et al. [190] investigated the stochastic bending–torsion coupled response of axially loaded slender composite beams with solid or thin-walled closed cross sections by using normal mode method in conjunction with receptance method. They used CBT with the effects of bending–torsion coupling and axial force included. Kiral [191] used a 3D FEM based on the classical laminated plate theory together with the Newmark integration method in order to obtain the dynamic response of the beam. He employed Rayleigh damping in the dynamic analyses and presented the impulse, step and moving load responses of the composite beam are for different damping ratios. Ibrahim et al. [192] investigated the periodic response of cross-ply composite curved beams subjected to harmonic excitation with frequency in the neighborhood of symmetric and anti-symmetric linear free vibration modes. Their analysis was carried out using HSDT based FEM.

6.2. Rotating beams

Rotating beams are mainly used in two applications. First, shafts that have tubular cross section and spin around longitudinal axis and second blades that usually have box cross section and rotate along normal axis. The first is mainly used in automotive industry and power transmission devices while the second have mainly aerospace applications.

6.2.1. Shafts

Singh et al. [193] reviewed the developments in dynamics of composite shafts in 1997. Singh and Gupta [194] investigated on natural frequencies and damping ratios in flexural modes. They used beam and shell formulation and concluded that their beam theory cannot account for bending–stretching and shear-normal couplings. Also for extreme values of length and thickness to radius ratio, the beam formulation was inaccurate. Kim and Bert [195] proposed a theoretical analysis for determining the shaft critical speeds using the dynamic analog of the Sanders best first approximation shell theory. They included the combined effects of torsion and rotational effects containing the centrifugal and Coriolis forces. Song and Librescu [196] studied the anisotropy and structural coupling effects on vibration and instability of spinning thin-walled beams. They included transverse shear and the primary and secondary warping effects. Kim et al. [197] investigated on free vibration of a rotating tapered composite Timoshenko shaft using Galerkin method. It was found that by tapering, bending natural frequencies and stiffness can be significantly increased over those of uniform shafts having the same volume and made of the same material.

Song et al. [198] investigated on vibration and stability of anisotropic pre-twisted beams rotating at constant angular speed. They used refined theory of thin-walled anisotropic composite beams featuring bending–bending elastic coupling. In a subsequent publication [199], they addressed problems related with the implications of conservative and gyroscopic forces on vibration and the stability of a circular cylindrical shaft modeled as a thin-walled spinning composite beam. Chang et al. [200] considered composite shaft containing discrete isotropic rigid disks and supported by bearings modeled as springs and viscous dampers. They extended Hamilton's principle to derive the governing equations for finding critical speeds of composite shaft systems and incorporated the transverse shear deformation, rotary inertia and gyroscopic effects, as well as the coupling effects due to the lamination of composite layers. Chang et al. [201] performed vibration analysis of rotating composite shafts containing randomly oriented reinforcements. Gubran and Gupta [202] analyzed the natural frequencies of composite shafts using equivalent modulus beam theory with shear

deformation, rotary inertia and gyroscopic effects included. Their approach took into account effects of stacking sequence and different coupling mechanisms. Banerjee and Su [80] developed the dynamic stiffness matrix of a spinning thin-walled composite beam and investigated its free vibration characteristics based on CBT. Na et al. [203] studied vibration and stability of a circular cylindrical shaft modeled as a tapered thin-walled composite beam, spinning with constant speed and subjected to an axial compressive force. Ghoneim and Lawrie [204] developed a mathematical model, based on Timoshenko beam assumption, for a rotating composite cylindrical shaft with cylindrical constrained layer damping partially covering the length span of the shaft. Sino et al. [205] introduced a homogenized FEM which takes into account internal damping of the beam and evaluated natural frequencies and instability thresholds of an internally damped rotating composite shaft. Qatu and Iqbal [21] used CBT to solve the vibration of a cross-ply laminated composite driveshaft with an intermediate joint. Alwan et al. [206] studied dynamic behavior of composite shafts with particular interest on estimation of damping. They used ANSYS for modeling of the shaft and analyzed the effect of material properties and stacking sequence on frequencies of composite tube-shafts. Different methodologies such as logarithmic decay curve, half-power method, and hysteresis loop method using force sensors were used for determining the damping of the composite shafts.

6.2.2. Blades

Kosmatka and Friedmann [95] determined the free vibration characteristics of composite turbo-propellers using a straight beam FEM. The FEM was obtained from Hamilton's principle, with allowances for generally anisotropic material behavior, arbitrary cross-sectional properties, large pre-twist angles, out-of-plane cross-section warping, and geometrically nonlinear behavior, based on moderate-deflection theory. Chandra and Chopra [141] presented a theoretical–experimental study of the free vibration characteristics of thin-walled composite box beams with bending–twist and extension–twist coupling under rotating conditions using Galerkin method. It was shown also that bending–shear coupling influences the flexural vibration frequencies of antisymmetric box beams significantly while extension–shear does not. Ozgumus and Kaya [69,70] performed free vibration analysis of a rotating, double tapered beam featuring coupling between flapwise bending and torsional vibrations using DTM. Chandiramani et al. [207] studied the free vibration behavior of rotating blades, modeled as hollow boxed beams. A nonlinear HSDT including warping inhibition and centrifugal force was developed. Kuang and Hsu [208] investigated on effect of fiber angle, internal and external damping, inclined angle and the rotation speed on the natural frequencies of orthotropic composite pre-twisted blades by employing the differential quadrature method (DQM). Choi et al. [169] studied bending vibration control of the pre-twisted rotating composite thin-walled beam. The formulation was based on single cell composite beam including a warping function, centrifugal force, Coriolis acceleration, pre-twist angle and piezoelectric effect. They used CBT along Kelvin–Voigt internal and linear external damping coefficients. Lee et al. [209,210] used Rayleigh–Ritz method to perform modal analysis of a rotating composite symmetric cantilever beam. Huang et al. [211] proposed a method based on the power series solution to solve the natural frequency of very slender rotating beam at high angular velocity. They investigated the natural frequency of the flapwise bending vibration, and coupled lagwise bending and axial vibration for the rotating beam. Arvin and Bakhtiari-Nejad [212] studied the linear and nonlinear (von-Karman) free vibrations of rotating composite Timoshenko beams. The prestressed configuration due to centrifugal forces was found using DTM. The direct multiple scales method was applied to the three-coupled

nonlinear updated partial differential equations for the nonlinear vibration study.

6.3. Damaged beams

Study on damaged beams is focused on two main areas. One aspect of research is to investigate the effect of damages on beams natural frequencies in order to avoid those ranges. Another aspect is to use vibration for finding damages present in the structure that is called vibration monitoring. Vibration monitoring is one of the main methods for damage identification and health monitoring of composite structures.

6.3.1. Damage effect on natural frequencies

Della and Shu [213] provided a relevant survey on various analytical models and numerical analyses for the free vibration of delaminated composites in 2007. Krawczuk and Ostachowicz [214] investigated modeling and vibration analysis of a cantilever composite beam with a transverse open crack. Gadelrab [97] used FEM to obtain the effect of the delamination length and its starting point from the end condition on the natural frequencies of composite laminated beams. Lee [59] used a layerwise theory for free vibration analysis of a laminated beam with delaminations. Lee et al. [215] made free vibration analysis of axially compressed laminated composite beam-columns with multiple delaminations. Birman and Byrd [216] investigated the effect of matrix cracks in longitudinal and transverse layers of cross-ply ceramic matrix composite beams on their mechanical properties and vibration frequencies. Kisa [217] used FEM and the component mode synthesis methods to perform free vibration analysis of a cantilever composite beam with multiple cracks. Ostachowicz and Zak [110] used HSDT based FEM to study damped vibration of a laminated cantilever beam with a single closing delamination. Perel [218] performed FSDT based FEM for vibration of delaminated composite beam with an account of contact of the delamination crack faces. Zak [219] studied damped nonlinear vibration of a delaminated composite beam using HSDT based FEM. Viola and Tornabene [220] studied in-plane free vibrations of thin and thick non-uniform circular arches in undamaged and damaged configurations with various boundary conditions. Structural damage was represented by one crack in different positions and with various damage levels. A cracked section was modelled as an elastic hinge with rotational constant and the equations of motion were solved using GDQ. Viola et al. [221] extended this study to multi-stepped and multi-damaged circular arches.

Della and Shu [222] presented an analytical solution for the free vibration of beams with two overlapping delaminations in pre-buckled states. They analyzed the delaminated beam as seven interconnected EB beams and observed a monotonic relation between the natural frequency and the compressive load. Baba and Gibson [223] used 2D FEA to predict the natural frequencies and corresponding vibration modes of a free-free sandwich beam with delamination of various sizes and locations. Baba and Thoppul [142] made a experimental study to determine the effect of curvatures and debond on the flexural stiffness and strength of composite sandwich beam structures. Kiral [224] studied the free vibration analysis of delaminated composite beams. Yazdi and Rezaeepazhand [225] studied the applicability of similitude theory in establishing necessary similarity conditions for designing scaled down models for predicting the vibration behavior of delaminated composite beam-plates. Kargarnovin et al. [226] studied the free vibration of generally laminated composite beams with a delamination. The constraint conditions were applied using Lagrange multipliers and the effects of material couplings with the effects of shear deformation, rotary inertia and Poisson's effect were taken into account.

6.3.2. Vibration monitoring

Zou et al. [227] conducted a review in 1998 on vibration based model independent damage identification and health monitoring of composite structures. Ratcliffe and Bagaria [228] presented an experimental nondestructive vibration-based technique for locating a delamination in a composite beam. Sahin and Sheno [229] investigated the effectiveness of the combination of global (changes in natural frequencies) and local (curvature mode shapes) vibration-based analysis data as input for artificial neural networks for location and severity prediction of damage in composite beams.

Ooijsavaar et al. [143] used experiments to investigate vibration based damage identification method for a 2.5-dimensional composite T-beam. Nichols and Murphy [230] worked on detecting delamination in composite beams based on a polyspectral analysis of the structure's vibrational response. They presented a low-dimensional model of the structure that captures the delamination-induced nonlinearity and showed how it influences the beam's dynamic response.

6.4. Added mass effect

Chandrashekhara and Bangera [231] investigated vibration of symmetrically laminated clamped-free beam with a mass at the free end. They derived equations of motion for the laminated beam accounting for the Poisson effect, rotary inertia and transverse shear deformation. White and Heppler [232] derived the equations of motion and boundary conditions for a free-free Timoshenko beam with rigid bodies attached at the endpoints and developed the frequency equation including the effects of the body mass, first moment of mass, and moment of inertia. Dadfarnia et al. [233] used a translational cantilevered EB beam with tip mass at its free end to study the effect of several damping mechanisms on the stabilization of the beam displacement. Specifically, a Lyapunov-based controller utilizing a partial differential equation model of the translational beam was developed to exponentially stabilize the beam displacement while the beam support was regulated to a desired set-point position. Eftekhari et al. [234] used the diversity guided evolutionary algorithm (DGEA) to find natural frequencies of a symmetrically laminated composite cantilever beam under base excitation with and without a tip mass. Also the multiple scales method was applied to determine the nonlinear response and stability of the system.

6.5. Damped and viscoelastic beams

Bishop and Kinra [235] studied on thermoelastic damping of a laminated beam in flexure and extension. Zapfe and Lesieutre [236] proposed a smeared laminate model for the dynamic analysis of laminated beams to predict the modal frequencies and damping of simply supported beams with integral viscoelastic layers. Their model included the effects of transverse shear and rotary inertia. In another research, they proposed a discrete layer beam FEM for the dynamic analysis of composite sandwich beams with integral damping layers [121]. Yim [237] compared three different methods for prediction of damping of a symmetric balanced laminated composite beam. Kovacs [238] proposed an iterative model to predict the modal frequencies and damping of simply-supported sandwich circular archs. He compared solutions for a three-layer circular arch with a three-layer approximate model.

Vengallatore [239] studied thermoelastic damping in symmetric, three-layered, laminated, micromechanical EB beams using an analytical framework developed by Bishop and Kinra [235]. Numayr and Qablan [240] considered three cases to analyze free vibrations of wide sandwich beams. They modeled viscoelastic core by elastic translational and rotational springs and used the finite difference method to solve the problem. They showed that if

the bending–torsion coupling was pronounced, the inclusion of warping affects the natural frequency considerably.

Saravanas et al. [127] developed 3D shear beam FEM for the damping analysis of tubular beams. Edery-Azulay and Abramovich [154] studied the effects of piezoceramic materials on the augmented damping of vibrating piezo-composite beams. Prabhakar and Vengallatore [241] presented an exact theory to compute the frequency dependence of thermoelastic damping in asymmetric, bilayered, micromechanical EB beam resonators. Vibration analysis of composite sandwich beams with viscoelastic core was investigated using DTM [75] and HSDT [242]. Alwan et al. [206] studied different methodologies such as logarithmic decay curve, half-power method, and hysteresis loop method using force sensors for determining damping of composite shafts.

6.6. Beams on elastic support

Patel et al. [117] studied nonlinear free flexural vibrations and post-buckling of laminated orthotropic beams resting on a class of two parameter elastic foundation using a three-noded shear flexible beam element. Their nonlinear formulation included the effects of transverse shear deformation, in-plane and rotary inertia terms. Koutsawa and Daya [243] performed static and free vibration analysis of laminated fiberglass beam on viscoelastic supports. Jafari-Talookolaei and Ahmadian [244] investigated free vibration analysis of a cross-ply laminated composite beam on two parameter elastic foundation (Pasternak) foundation. They computed natural frequencies of beam on Pasternak foundation using FEM on the basis of Timoshenko beam theory. Malekzadeh and Vosoughi [245] used DQM for large amplitude free vibration analysis of laminated composite thin beams on nonlinear elastic foundation. Beam edges were considered to be elastically restrained against rotation and in-plane immovable. They developed a FEM program to verify the results of the presented DQM approach. Baghani et al. [246] presented analytical expressions for large amplitude free vibration and post-buckling analysis of unsymmetrically laminated composite beams on elastic foundation. They solved nonlinear governing equation by employing the variational iteration method and reached accurate solutions for a wide range of vibration amplitudes.

6.7. Other complexities

Vibration analysis of stepped laminated composite Timoshenko beams was investigated by Farghaly and Gadelrab [247] and Dong et al. [248]. They determined flexural rigidity and transverse shear rigidity of a laminated beam based on FSDT. Kütüg [249] investigated the frequency and forms of natural vibrations of a hinged beam-strip fabricated from a composite material with small-scale curved structures using the plate theory in the framework of the Timoshenko hypothesis. Nachum and Altus [250] studied natural frequencies and mode shapes of nonhomogeneous (deterministic and stochastic) rods and beams based on the functional perturbation method.

7. Concluding remarks

Researches on the vibration and dynamic analyses of composite beams have been increasing rapidly in the last two decades. A bibliography of 250 research articles has been cited on the subject during this time with more than two third in the last 12 years. Researchers in the field of composite beam analyses avoided use of 3D theory of elasticity and developed and used thick beam theories. Many researchers use the finite element method for their analyses in different aspects such as smart or damaged beams. Much of the research is paying attention to evolving technologies

on smart materials, and applications such as blades and shafts. Other research focused on vibration control through damping and structural health monitoring through vibration testing.

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